



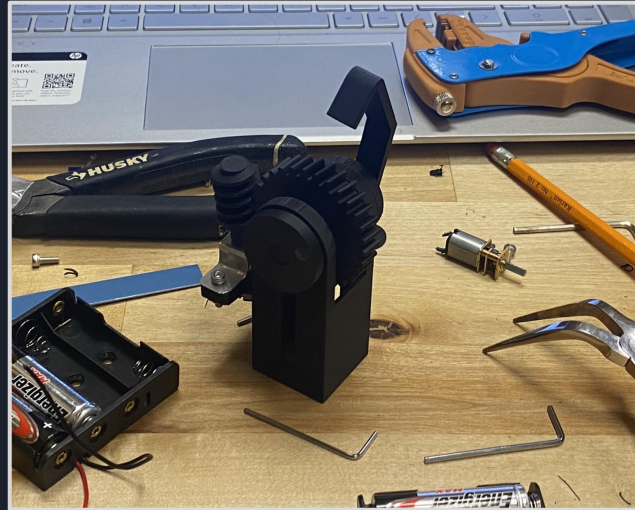
Children's Toy Prototype Overview

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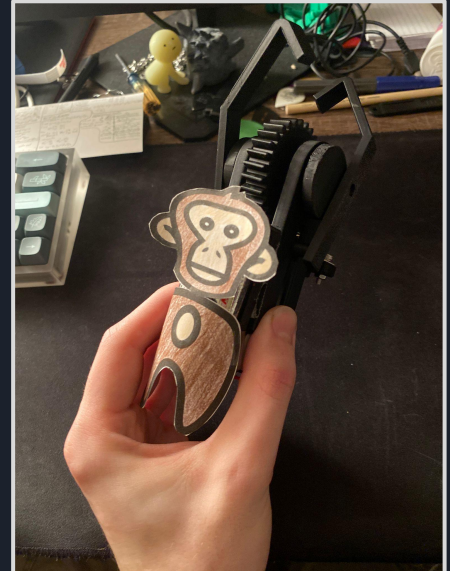
Physical Prototype



Video demonstration of product's functionality



Assembly process featuring gear box and internal mechanisms



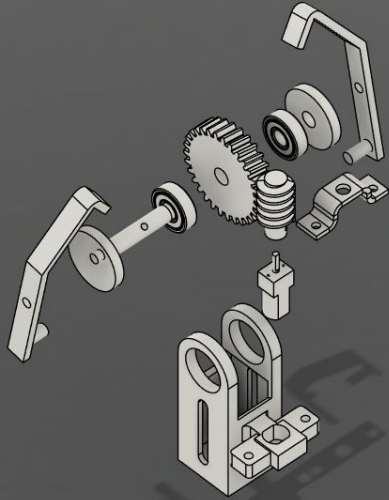
Final prototype of product with finished design.



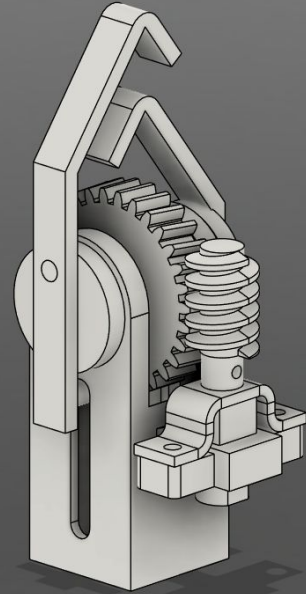
Design Priorities

- **Safety:**
 - During our planning process, the safety of our users was one of the most important objectives. This resulted in minimizing the size and weight of our design. The final prototype achieved a final weight under 700g and a manageable body size of 2.38"x 2.03"x 5.01".
- **Reliability:**
 - Consistent functionality was another significant aim for our finished product. We designed our robot to be slow and methodical to maximize the design's success. This constraint resulted in a completion across the rope of around 2 minutes.
- **Ease of use:**
 - Another important feature of our product was the ability for a person of any age to operate the toy without harm and external assistance. This criteria caused us to design a mechanism to secure the rope, as well as a simple activation switch.

CAD Design

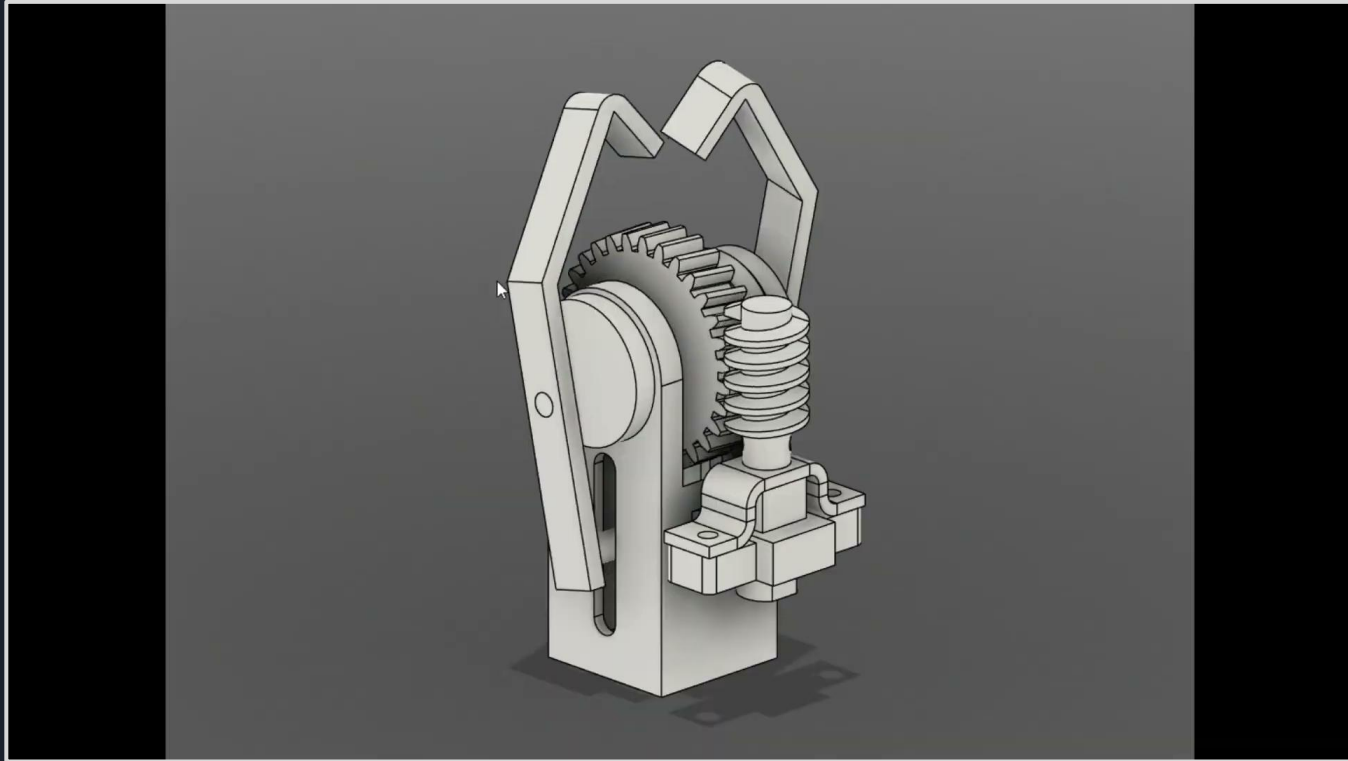


Exploded View of Final Prototype



Assembled View of Final Prototype

CAD Motion Demonstration





Cost Evaluation

- The body, gears, and arms are produced in-house via injection molding
- Off-the-shelf parts include:
 - Motor: \$12.50
 - Battery holder/wiring: \$4.25
 - Switch: \$0.35
 - Fasteners: \$0.08

Item	One-Time tooling	Per-Unit cost	Total run cost
Body	\$8,000.00	\$0.54	\$10,700.00
Worm Gears	\$6,000.00	\$0.16	\$6,800.00
Arms / body decoration	\$5,000.00	\$0.25	\$6,200.00
Off-the-shelf cost	N/A	\$17.42	\$62,500.00
Total of All parts (Per 5,000 units)		\$18.32	\$110,600.00



Sales Pitch

Investing Promises:

- **Strong Industrial Logistics:**
 - Our design is very cheap to manufacture, allowing for high profit margins. In addition to that, its overall size is small compared to other toys of this category. It is also easy to package/transport considering it has a box-like design and removable limbs. All these advantages compound together, making manufacturing and delivery low cost and effective.
- **Appeal to Parents:**
 - Parents are the ultimate decision makers when it comes to what toys their kids have access to. So, it's important we appeal to their needs. The toy is simple yet exposes the kid to dynamic movements that mimic nature accurately. Exposure to these things is important in child development, but it cannot be too complicated. Which is why our design's mechanics are digestible and engaging for children.
- **Appeal to Young Users:**
 - A kid's experience and excitement for a toy heavily influences their parent's decision to buy it, and our design does not fall short in that department. Our toy's action is unique and fascinating to kids. Especially to those who have an interest in animals. On top of that, it is very easy to use. Attaching the toy to a string is simple and turning it on is done with a basic switch.



Thank You